

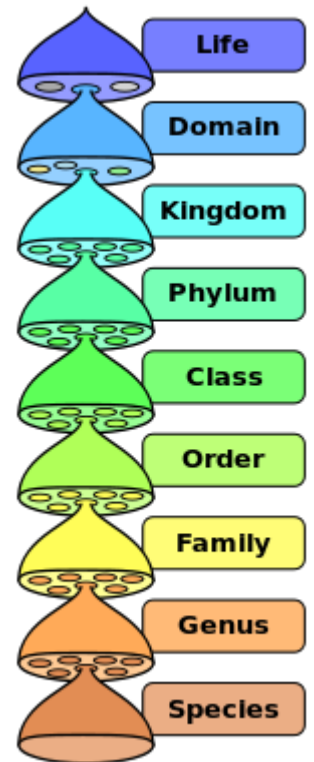
Species

In biology, a **species** (/ˈspiːʃiːz/ [ⓘ] [ⓘ]listen)) is the basic unit of classification and a taxonomic rank of an organism, as well as a unit of biodiversity. A species is often defined as the largest group of organisms in which any two individuals of the appropriate sexes or mating types can produce fertile offspring, typically by sexual reproduction. Other ways of defining species include their karyotype, DNA sequence, morphology, behaviour or ecological niche. In addition, paleontologists use the concept of the chronospecies since fossil reproduction cannot be examined.

All species (except viruses) are given a two-part name, a "binomial". The first part of a binomial is the genus to which the species belongs. The second part is called the specific name or the specific epithet (in botanical nomenclature, also sometimes in zoological nomenclature). For example, *Boa constrictor* is one of four species of the genus *Boa*.

While the definitions given above may seem adequate, when looked at more closely they represent problematic species concepts. For example, the boundaries between closely related species become unclear with hybridisation, in a species complex of hundreds of similar microspecies, and in a ring species. Also, among organisms that reproduce only asexually, the concept of a reproductive species breaks down, and each clone is potentially a microspecies. Though none of these are entirely satisfactory definitions, scientists and conservationists need a species definition which allows them to work, regardless of the theoretical difficulties. If species were fixed and clearly distinct from one another, there would be no problem, but evolutionary processes cause species to change continually, and to grade into one another.

Species were seen from the time of Aristotle until the 18th century as fixed categories that could be arranged in a hierarchy, the great chain of being. In the 19th century, biologists grasped that species could evolve given sufficient time. Charles Darwin's 1859 book *The Origin of Species* explained how species could arise by natural selection. That understanding was greatly extended in the 20th century through genetics and population ecology. Genetic variability arises from mutations and recombination, while organisms themselves are mobile, leading to geographical isolation and genetic drift with varying selection pressures. Genes can sometimes be exchanged between species by horizontal gene transfer; new species can arise rapidly through hybridisation and polyploidy; and species may become extinct for a variety of reasons. Viruses are a special case, driven by a balance of mutation and selection, and can be treated as quasispecies.



The hierarchy of biological classification's eight major taxonomic ranks. A genus contains one or more species. Intermediate minor rankings are not shown.

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